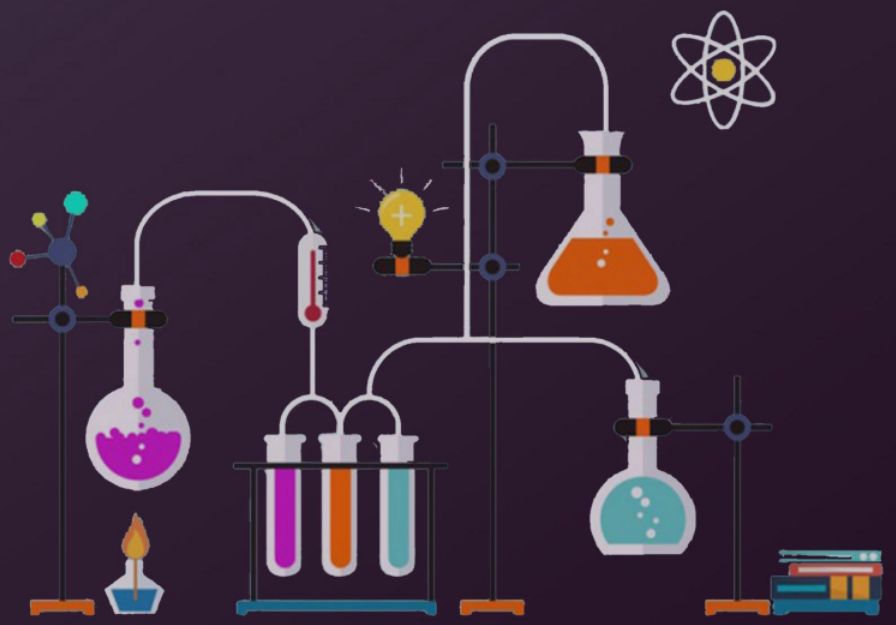
CHEMISTRY

d and f Block



SHEET

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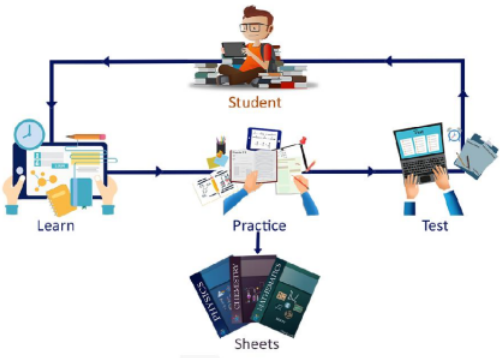
Contents

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	General Trends in d-block Elements	1-7	1,3-9,15, 16,19,20	1-3	1-3,9-12,22
2	Colour, Interstitial Compounds And Alloys of Transition Metals	8	2,21	4	4
3	Oxides and Oxoanions of d-block Metals	9-13	13,17	5	24
4	Lanthanides	14-17	18	6-13	14,15
5	ACC	18-22	10-12,14	14-16	5-8,13, 16-21,23,25

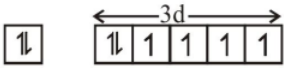
Exercise - 1 — Concept Builder

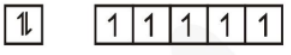
SINGLE CORRECT TYPE

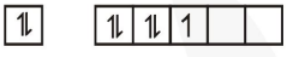
*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

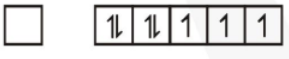
General trends in D-block elements

Q1. The manganese (Z = 25) has the outer configuration:-

(A) 

(B) 

(C) 

(D) 

Q2. In the first transition series the melting point of Zn is low, because :-

(A) Metallic bonds are strong due to d^{10} configuration

(B) Metallic bonds are weak due to d^5 configuration

(C) Metallic bonds are weak due to d^7 configuration

(D) d-orbitals have no unpaired electrons

Q3. Which element is having lowest melting and boiling point :-

(A) Ti (B) Cu

(C) Zn (D) Mn

Q4. The most stable oxidation state of Mn is.

(A) $Mn^{+4}(d^3)$ (B) $Mn^{+6}(d^1)$\

(C) $Mn^{+2}(d^5)$ (D) $Mn^{+3}(d^4)$

Q5. Which of the following element have maximum density :-

(A) Sc (B) Co

(C) Fe (D) Mn

Q6. Which aquated ion is colored :-

(A) Zn^{+2} (B) Cu^{+}

(C) Sc^{+3} (D) None of these

Q7. In first transition series, the melting point of Mn is low because :-

(A) Due to d^{10} configuration, metallic bond are strong

(B) Due to d^7 configuration, metallic bond are weak

(C) Due to d^5 configuration metallic bonds are weak

(D) None of the above

Colour, Interstitial Compounds and Alloys of Transition Metals

Q8. Which of the following compounds is most sensetive of light ?

(A) AgCl (B) AgCN

(C) AgI (D) AgBr

Oxides and Oxoanions of d-block Metals

Q9. What would happen when a solution of potassium chromate is treated with an excess of dilute nitric acid -

(A) Cr^{3+} and $Cr_2O_7^{2-}$ are formed

(B) $Cr_2O_7^{2-}$ and H_2O are formed

(C) $Cr_2O_7^{2-}$ is reduced to +3 state of Cr

(D) $Cr_2O_7^{2-}$ is oxidised to +7 state of Cr

Q10. In dichromate dianion :-

(A) 4 Cr–O bonds are equivalent

(B) 6 Cr–O bonds are equivalent

(C) All Cr–O bond are equivalent

(D) All Cr–O bond are non equivalent

Q11. Which of the following exhibit colour due to charge transfer phenomenon, but not due to d-d transition ?

(A) $Cr_2O_7^{2-}$

(B) MnO_4^-

(C) CrO_4^{2-}

(D) All of these

Q12. Which of the following oxides is basic?

(A) Mn_2O_3

(B) MnO

(C) MnO_2

(D) Mn_2O_7

Q13. Deep violet color of $KMnO_4$ is due to :-

(A) d-d transition

(B) Charge transfer spectra

(C) d-f transition

(D) Nuclear charge

Lanthanides

Q14. The correct order of ionic radii of Y^{3+}, La^{3+}, Eu^{3+} and Lu^{3+} is :-

(A) $Y^{3+} < La^{3+} < Eu^{3+} < Lu^{3+}$

(B) $Y^{3+} < Lu^{3+} < Eu^{3+} < La^{3+}$

(C) $Lu^{3+} < Eu^{3+} < La^{3+} < Y^{3+}$

(D) $La^{3+} < Eu^{3+} < Lu^{3+} < Y^{3+}$

Q15. Which of the following shows maximum +8 oxidation state ?

(A) Re

(B) Os

(C) W

(D) Ir

Q16. 5f-subshell is filled by electron(s) -

(A) In actinoids

(B) After filling of 7s-subshell

(C) Before filling of electron in 6d series

(D) All are correct

Q17. Last element of Lanthanoid series is -

(A) Lawrencium

(B) Lutetium

(C) Thulium

(D) Hafnium

ACC

Q18. $Cr_2O_7^{2-} \underset{Y}{\overset{X}{\rightleftharpoons}} 2CrO_4^{2-}$, X and Y are respectively -

(A) $X = OH^-$, $Y = H^+$

(B) $X = H^+$, $Y = OH^-$

(C) $X = OH^-$, $Y = H_2O_2$

(D) $X = H_2O_2$, $Y = OH^-$

Q19. In which of following substance on heating change their colour as well as their oxidation state of metal.

(A) ZnO

(B) PbO

(C) HgI_2

(D) CrO_3

Q20. $N_2(g) + 3H_2(g) \overset{Fe+Mo}{\rightleftharpoons} 2NH_3(g)$; Haber's process, Mo is used as :

(A) A catalyst

(B) A catalytic promoter

(C) An oxidising agent

(D) As a catalytic poison

Q21. Manganese steel is used for making railways because **:**

(A) It is hard with high percentage of Mn

(B) It is soft with high percentage of Mn

(C) It is hard with small concentration of manganese with the impurities

(D) It is soft with small concentration of manganese with impurities.

Q22. Which is consequence of Lanthanoid contraction -

(A) Size of Zr >> Hf

(B) Size of Zr << Hf

(C) Size of Zr ≈ Hf

(D) Size of Zr > Zr^{4+}

Exercise - 1A — Concept Builder

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The highest oxidation state shown by transition element is :

(A) + 7 by Mn (B) + 8 by Os
(C) + 8 by Ru (D) + 7 by Fe

Q2. Which pair of ions is coloured :

(A) Mn^{+3}, Co^{+3} (B) Fe^{3+}, Cr^{+3}
(C) Zn^{2+}, Sc^{3+} (D) Ti^{2+}, Cu^{2+}

Q3. Cu is slowly oxidised in moisture to 'X', often seen in electrical circuits.

Select correct option(s) :-

(A) X is $Cu(OH)_2 \cdot CuCO_3$
(B) X is Blue substance
(C) Process is example of corrosion.
(D) X is $CuSO_4 \cdot 5H_2O$

Q4. Au + aqua regia → 'X'. About 'X' which of the following is/are incorrect?

(A) 'X' is $H[AuCl_4]$ and a square planar complex.
(B) X is $H[AuCl_4]$ and a tetrahedral complex.
(C) X is $AuCl_3$ and a triangular planar compound.
(D) X is $H_3[AuCl_6]$ and octahedral complex.

Q5. A black oxide of formula AgO is formed by strong oxidation of Ag_2O in alkaline solution.

Select correct option(s):-

(A) AgO is paramagnetic because it has unpair electron.
(B) AgO is diamagnetic because it has no unpair electron.
(C) AgO is diamagnetic because it contains both Ag^+ and Ag^{3+}
(D) AgO is acidic oxide of silver.

Paragraph for Questions 6 to 9

Transition metal and their compounds are used as catalysts in industry and in biological system. For example, in the Contact process, vanadium compounds in the +5 state (V_2O_5 or VO_3^-) are used to oxidise SO_2 to SO_3:

$$SO_2 + \frac{1}{2}O_2 \xrightarrow{V_2O_5} SO_3$$

It is thought that the actual oxidation process takes place in two stages. In the first step, V^{5+} in the presence of oxide ions converts SO_2 to SO_3. At the same time, V^{5+} is reduced to V^{4+}.

$$2V^{5+} + O^{2-} + SO_2 \longrightarrow 2V^{4+} + SO_3$$

In the second step, V^{5+} is regenerated from V^{4+} by oxygen :

$$2V^{4+} + \frac{1}{2}O_2 \longrightarrow 2V^{5+} + O^{2-}$$

The overall process is, of course, the sum of these two steps:

$$SO_2 + \frac{1}{2}O_2 \longrightarrow SO_3$$

Q6. Transition metals and their compounds catalyse reactions because :

(A) They have competely filled s-subshell
(B) They have a comparable size due to poor shielding of d-subshell
(C) They introduce an entirely new reaction mechanism with a lower and higher activation energy
(D) They have variable oxidation states.

Q7. During the course of the reaction :

(A) Catalyst undergoes changes in oxidation state.
(B) Catalyst increases the rate constant.
(C) Catalyst is regenerated in its orginal form when the reactants form the products.
(D) All are correct.

Q8. Catalytic activity of transition metals depends on :

(A) Their ability to exist in different oxidation states

(B) The size of the metal atoms

(C) The number of empty atomic orbitals available

(D) None of these

Q9. Which of the following ion involved in the above process will show paramagnetism?

(A) V^{5+}

(B) V^{4+}

(C) O^{2-}

(D) VO_3^-

Paragraph for Q. 10 to 12

Copper is the most noble of the first row transition elements. It occurs in small deposits in several countries. Ores of copper include chalcanthite ($CuSO_4 \cdot 5H_2O$),. atacamite [$Cu_2Cl(OH)_3$], cuprite (Cu_2O), copper glance (Cu_2S), and malachite [$Cu_2(OH)_2CO_3$]. However, 80% of the world copper production comes from the ore chalcopyrite ($CuFeS_2$). Extraction of copper from chalcopyrite includes roasting, iron removal, and self-reduction.

Q10. Partial roasting of chalcopyrite produces:

(A) Cu_2S and FeO (B) Cu_2O and FeO

(C) CuS and Fe_2O_3 (D) Cu_2O and Fe_2O_3

Q11. Iron is removed from chalcopyrite as:

(A) FeO (B) FeS

(C) Fe_2O_3 (D) $FeSiO_3$

Q12. In self-reduction, the reducing species is:

(A) S (B) O^{2-} (C) S^{2-} (D) SO_2

MATCHING LIST TYPE

Each question has TWO (02) ***matching lists: LIST I and LIST II****. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q13.

List-I	**List-II**
(P) Cr_2O_3	(1) Neutral oxide
(Q) CrO_3	(2) Amphoteric oxide
(R) Fe_3O_4	(3) Mix oxide
(S) N_2O	(4) Acidic oxide

Select correct code for matching -

Code :

	P	Q	R	S
(A)	2	4	3	1
(B)	2	3	4	1
(C)	4	2	3	1
(D)	4	3	1	2

Q14. Match the following column I and II :-

List-I	**List-II**
(P) Gun metal	(A) Cu +Sn
(Q) Solder	(B) Cu + Zn + Ni
(R) German silver	(C) Cu + Sn + Zn
(S) Bell metal	(D) Pb + Sn

(A) (P-A), (Q-C), (R-D), (S-B)

(B) (P-C), (Q-D), (R-B), (S-A)

(C) (P-C), (Q-B), (R-D), (S-A)

(D) (P-B), (Q-A), (R-C), (S-D)

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q15. Match the compounds of column X with oxidation state of column - II.

Column - I	**Column - II**
(A) $[Cr(H_2O)_6]Cl_3$	(P) 5
(B) CrO_5	(Q) 4
(C) K_3CrO_8	(R) 6
	(S) 3

ASSERTION AND REASON

(A) Statement-I is true,Statement-II is true; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I.

(C) Statement-I is true, Statement-II is false.

(D) Statement-I is false, Statement-II is true.

Q16. **Statement-I :** Tungsten has a very high melting point.

Statement-II : Tungsten is a covalent compound.

Q17. **Statement-I :** $KMnO_4$ is purple in colour due to charge transfer.

Statement-II : In MnO_4^-, there is no electron present in d-orbitals of manganese.

Q18. **Statement-I :** Separation of Zr and Hf is difficult.

Statement-II : Because Zr and Hf lie in the same group of the periodic table.

SUBJECTIVE TYPE

Q19. Why first ionisation energy of Cu is higher than that of sodium ?

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q20. The oxidation number of Mn in the product of alkaline oxidative fusion of MnO_2 is _____.

Q21. Number of ions which gives blue colour in aqueous state :

V^{+4}, Ni^{+2}, Ti^{+3}, Co^{+2}, Fe^{+3}, Cu^{+2}

Exercise - 2 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. d-block metals exhibit a general electronic configuration of -
(A) $(n-1)\ s^2p^6ns^1$
(B) $(n-1)s^2p^6d^5ns^1$
(C) $(n-1)s^2p^6ns^2np^1$
(D) $(n-1)s^2p^6d^{1-10}ns^{0-2}$

Q2. If the magnetic moment of Ti in a compound is 1.73 BM then oxidation state of Ti in that compound :
(A) +1 (B) +4
(C) +3 (D) +2

Q3. Which group contains coloured ions–
(A) Ti^{+4}, Cu^{+} (B) V^{+3}, Ti^{+3}
(C) Sc^{+3}, Zn^{+2} (D) La^{+3}, Hg^{+2}

Q4. Electronic configuration of a transition element X in +3 oxidation state is [Ar] $3d^5$ and Y in +2 state is [Ar] $3d^8$. What are the atomic number of the elements ?
(A) 23 and 26
(B) 26 and 28
(C) 21 and 25
(D) 24 and 26

Q5. Which does not decolourise $KMnO_4$ aqueous solution :-
(A) $C_2O_4^{2-}$ (B) HSO_3^-
(C) CO_3^{2-} (D) SO_3^{2-}

Q6. If compound absorbs violet color it appears in which color :-
(A) Green (B) Red
(C) Blue (D) Yellow

Q7. Deep violet color of $KMnO_4$ is due to :-
(A) d-d transition
(B) Charge transfer spectra
(C) d-f transition
(D) Nuclear charge

Q8. $KI + KMnO_4 \xrightarrow[\text{Weak alkaline medium}]{} MnO_2 + \text{'x'}$
Product 'x' may be :-
(A) Iodine
(B) Potassium iodate
(C) Potassium Per iodate
(D) Potassium Hypo iodate

Q9. The decreasing order of proton number in Y^{3+}, La^{3+}, Eu^{3+} and Lu^{3+} is :-
(A) $Y^{3+} < La^{3+} < Eu^{3+} < Lu^{3+}$
(B) $Y^{3+} > Lu^{3+} > Eu^{3+} > La^{3+}$
(C) $Lu^{3+} < Eu^{3+} < La^{3+} < Y^{3+}$
(D) $La^{3+} < Eu^{3+} < Lu^{3+} < Y^{3+}$

Q10. The reason for the stability of Gd^{3+} ion is -
(A) $4f$ subshell — half filled
(B) $4f$ subshell — completely filled
(C) Possesses the general electronic configuration of noble gases
(D) $4f$ subshell empty

Q11. Which of the following exhibits only +3 oxidation state ?
(A) Ac (B) Pa
(C) U (D) Th

Q12. Which of the following actinoids show oxidation states upto +7?
(A) Am (B) Pu
(C) U (D) Np

Q13. A reduction in atomic size with increase in atomic number is a characteristic of elements of :
(A) f-Block
(B) Radioactive series
(C) High atomic masses
(D) d-Block

Q14. Which of the following is correct statement :-
(A) Cu^{2+} is more stable than Cu^{+}
(B) Ti^{4+} is more stable than Ti^{3+}
(C) Mn^{2+} is more stable than Mn^{3+}
(D) All are correct

Q15. Magnetic moment of x^{n+} is $\sqrt{24}$ B.M. Hence No. of unpaired electron and value of 'n' respectively. (Atomic number = 26)
(A) 4 , 3 (B) 3 , 5
(C) 4 , 2 (D) 4 , 1

Q16. Select the correct order :-
(A) $Co^{+3} < Fe^{+3}$ (Paramagnetic property)
(B) Cu < Zn (density)
(C) Ni < Cu > Zn (size)
(D) All of them

Exercise - 2A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Q1. An element of 3d-transition series shows two oxidation states x and y, differ by two units then
(A) compounds in oxidation state x are ionic if x > y
(B) compounds in oxidation state x are ionic if x < y
(C) compounds in oxidation state y are covalent if x < y
(D) compounds in oxidation state y are covalent if y < x

Q2. Amongst CuF_2, $CuCl_2$ and $CuBr_2$
(A) only CuF_2 is ionic
(B) both $CuCl_2$ and $CuBr_2$ are covalent
(C) CuF_2 and $CuCl_2$ are ionic but $CuBr_2$ is covalent
(D) CuF_2, $CuCl_2$ as well as $CuBr_2$ are ionic

Q3. Which of the following statement(s) is(are) correct with reference to the ferrous and ferric ions ?
(A) Fe^{3+} gives brown colour with potassium ferricyanide
(B) Fe^{2+} gives blue precipitate with potassium ferricyanide
(C) Fe^{3+} gives red colour with potassium thiocyanate
(D) Fe^{2+} gives brown colour with ammonium thiocyanate

Q4. Which of the following statements concern with transition metals?
(A) compounds containing ions of transition elements are usually coloured
(B) the most common oxidation state is +3
(C) they show variable oxidation states, which differ by two units only
(D) they easily form complexes

Q5. To an acidified dichromate solution, a pinch of Na_2O_2 is added and shaken. What is observed:
(A) blue colour
(B) Orange colour changing to green
(C) Copious evolution of oxygen
(D) Bluish - green precipitate

Q6. $CuSO_4(aq) + 4NH_3 \longrightarrow X$, then X is
(A) $[Cu(NH_3)_4]^{2+}$
(B) paramagnetic
(C) coloured
(D) of a magnetic moment of 1.73 BM

Q7. Correct statement(s) is/are
(A) an acidified solution of $K_2Cr_2O_7$ liberates iodine from KI
(B) $K_2Cr_2O_7$ is used as a standard solution for estimation of Fe^{2+} ions
(C) in acidic medium, M = N/6 for $K_2Cr_2O_7$
(D) $(NH_4)_2Cr_2O_7$ on heating decomposes to yield Cr_2O_3 through an exothermic reaction

Q8. In the equation :

$M+8CN^- + 2H_2O + O_2 \rightarrow 4[M(CN)_2]^- + 4OH^-$

metal M is

(A) Ag (B) Au

(C) Cu (D) Hg

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 9 to 12

The elements of the three transition series of the d-block are given below:

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Y	Zr	Bb	Mo	Tc	Ru	Rh	Rd	Ag	Cd
La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg

In any transition series, as we move form left to right, the d-orbitals are progressively filled and their properties vary accordingly.

Q9. Which element do you expect to have the smallest atomic radius ?

(A) Sc (B) Zn

(C) La (D) Hg

Q10. Which element do you expect to have the highest melting point?

(A) La (B) W

(C) Os (D) Pt

Q11. Which element out of the following do you expect to have the lowest melting point?

(A) Cr (B) Mn

(C) Fe (D) Co

Q12. Which of the following is the correct order of second ionization energy?

(A) V > Cr > Mn

(B) V < Cr < Mn

(C) V < Cr > Mn

(D) V > Cr < Mn

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q13.

List-I **(Metal ion of 3d-series)**	**List-II** **(Characterstic)**
(P) Ni^{2+}	(1) produce blue aq. solution
(Q) Cr^{2+}	(2) half field t_{2g} in octahedral complex
(R) V^{2+}	(3) diamagnetic ion
(S) Ti^{4+}	(4) calculated μ = 2.84 B.M. (spin only)

Select correct code for matching -

Code :

	P	Q	R	S
(A)	4	2	1	3
(B)	3	1	2	4
(C)	4	1	2	3
(D)	1	2	4	3

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q14. Match the compounds/elements given in Column I with uses given in Column II.

Column I (Compound/element)	**Column II (Use)**
(A) Lanthanoid oxide	(P) Production of iron alloy
(B) Lanthanoid	(Q) Television screen
(C) Misch metal	(R) Petroleum cracking
(D) Magnesium based alloy is	(S) Constituent of (Lanthanoid metal + iron)
(E) Mixed oxides of lanthanoids	(T) Bullets lanthanoids are employed
	(U) In X-ray screen

Q15. Match the property given in Column I with the element given in Column II.

Column I (Property)	**Column II (Element)**
(A) Lanthanoid which shows +4 oxidation state	(P) Pm
(B) Lanthanoid which can show +2 oxidation state	(Q) Ce
(C) Radioactive lanthanoid	(R) Lu
(D) Lanthanoid which has $4f^7$ electronic configuration in +3 oxidation state	(S) Eu
(E) Lanthanoid which has $4f^{14}$ electronic configuration in +3 oxidation state	(T) Gd
	(U) Dy

Q16. **Match the column :-**

Column - I (Property)	**Column - II (Transition elements)**
(A) Highest Oxidation state	(P) Cr
(B) Highest density	(Q) Os
(C) Element with maximum unpaired electrons	(R) Tc
(D) Radioactive transition element	(S) Ru

TRIPLE MATCH

*This section contains TWO tables (each having 3 columns and 4 rows). Based on each table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Column-I contains reactions, Column-II contains one of the products of reaction and Column III contains nature of same product.

Column-I	**Column-II**	**Column-III**
(A) Cu + dil. HNO_3	(P) NO	(X) Neutral diamagnetic gas
(B) Cu + Conc. HNO_3	(Q) NO_2	(Y) Paramagnetic neutral gas
(C) Zn + dil. HNO_3	(R) N_2O	(Z) Acidic paramagnetic gas
(D) Zn + Conc. HNO_3	(S) N_2O_3	(W) Blue solid

Q17. Which of the following options is the correct combination?

(A) -(A)(P)(Y) (B) -(A)(Q)(R)
(C) -(C)(P)(Z) (D) -(B)(Q)(R)

Q18. Which of the following options is the correct combination for brown gas formed as product?

(A) -(D)(S)(Z) (B) -(D)(Q)(Z)
(C) -(A)(S)(Z) (D) -(B)(Q)(R)

Q19. Select the correct option which is combination for linear shape of gas formed as product?

(A) -(A)(R)(X) (B) -(D)(P)(Y)
(C) -(A)(S)(Y) (D) -(C)(R)(X)

ASSERTION AND REASON

(A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.

(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for Statement-I.

(C) Statement-I is true, Statement-II is false.

(D) Statement-I is false, Statement-II is true.

Q20. **Statement-I :** K_2CrO_4 has yellow colour due to charge transfer.

Statement-II : CrO_4^{2-} ion is tetrahedral in shape.

Q21. **Statement-I :** Cu^+ ion is colourless.

Statement-II : Four water molecules are coordinated to Cu^+ ion in water.

SUBJECTIVE TYPE

Q22. Of the two common oxides of chromium, chromium (VI) oxide and chromium (III) oxide, which should have the lower melting point? Explain your reasoning.

Q23. Identify (A) to (F) in the following :

$FeSO_4$ (A) $\xrightarrow{\Delta}$ (B) + (C) + (D) (C) and (D) are gases, (B) is red-brown solid, (C) can be oxidised to (D). (C) also turns $Cr_2O_7^{2-}/H^+$ solution green. (B) dissolves in HCl forming deep yellow solution (E). (E) gives blue colour (F) with $K_4[Fe(CN)_6]$. (D) dissolves in H_2O forming a well knows mineral acid called "king of acids".

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q24. Define the oxidation states of Mn in product of the given reaction

$3K_2MnO_4+2H_2O+4CO_2 \rightarrow 2X+Y+4KHCO_3$

If the oxidation state of Mn in product X and Y are n_1 and n_2 respectively.

Then find out the value of (n_1+n_2)

Q25. The number of water molecule(s) directly bonded to the metal center is $CuSO_4 \cdot 5H_2O$ is _____.

Answer Key

Ex - 1									
1	2	3	4	5	6	7	8	9	10
B	D	C	C	B	D	C	D	B	B
11	12	13	14	15	16	17	18	19	20
D	B	B	B	B	D	B	A	D	B
21	22								
A	C								

Ex-1A

1. A,B,C **2.** A,B,D **3.** A,B,C **4.** B,C,D **5.** A,D

6. D **7.** D **8.** A **9.** B **10.** A

11. D **12.** C **13.** A **14.** B

15. (A) → (S); (B) → (R); (C) → (P)

16. C **17.** B **18.** B **20.** 6 **21.** 2

Ex - 2									
1	2	3	4	5	6	7	8	9	10
D	C	B	B	C	D	B	B	B	A
11	12	13	14	15	16				
A	B,D	A	D	C	A				

Ex-2A

1. B,C **2.** A, B **3.** A,B,C **4.** A, B, D **5.** A, C

6. A, B, C, D **7.** A, C, C **8.** A, B **9.** B **10.** B

11. B **12.** C **13.** C

14. (A) → (Q) ; (B) → (P) ; (C) → (S) ; (D) → (T) ; (E) → (R)

15. (A) → (Q) ; (B) → (S) ; (C) → (P) ; (D) → (T) ; (E) → (R)

16. (A) → (Q),(S) ; (B) → (Q) ; (C) → (P) ; (D) → (R)

17. A **18.** B **19.** D **20.** B **21.** C

23. A = $FeSO_4$, B = Fe_2O_3, C = SO_2, D = SO_3, E = $FeCl_2$, F = $Fe_2[Fe(CN)_6]$

24. 11 **25.** 4

-: Important Notes & Formulas :-